

The Algebraic Cycle as a Tail Witness

The Hodge Verdict Is Not Frame-Internal

The circumscription obstruction in algebraic geometry, the inverted witness asymmetry, the confirmation of the Compiled Edition's instrument theorems as circumscription theorems, and the completed spine

Lu Semita · EmergenceByDesign Drawing definitively on the prior work: the Hodge Traversal Compiled Edition V4–V8 (whose instrument theorems this paper shows to be exactly the circumscription theorems of the T-series), the Obstruction Calculus Consolidation (C1), the Residue Thesis (R1), and the companion papers T1 (Navier–Stokes), T2 (Riemann), and T3 (Yang–Mills), whose tetralogy this paper completes.

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Tier discipline throughout: **Tier I** is standard checkable mathematics, asserted with complete proof or standard citation; **Tier II** is exact structure read after the Tier-I objects are fixed; **Tier III** is declared direction. Falsification surfaces in §9.

0. Origin

The tetralogy closes where the program's discipline was forged. The originating intuition, stated as in T1–T3:

Each of these questions is of the same form — the way the question is posed is what generates the outcome. One must arbitrarily stop imposing the same circumscription continuum to assert a definitive about what is unwitnessable from its own framing. The issue is not domain-dependent; it is structural.

The Hodge conjecture occupies a special position in this program: it is the domain in which the traversal method was first executed to completion (the Compiled Edition V4–V8), in which the obstruction calculus found its flagship instantiation (C1, Theorems 4.3–4.6), and — decisively — in which the structural reading received its historical validation *during the program's own lifetime*: the Weil fourfold frontier, which the instrument proved undecidable by every frame-internal operation, was crossed by Markman through exactly the door, by exactly the weld type, at exactly the coordinates the instrument had posted. No other domain in the tetralogy offers a documented, dated event in which the circumscription analysis predicted the shape of its own supersession and was confirmed.

This paper therefore has a task the other three did not: not to construct the circumscription theorems for a new domain, but to **prove an identity** — that the Compiled Edition's instrument theorems, quantifier exhaustion, verdict lattice, and door pricing *are* the T-series theorems, stated before the T-series existed; that the Hodge obstruction is the same structural obstruction, rigorously and definitively; and that with this identification the spine of §5 closes over all four domains with no remainder. Along the way the Hodge case contributes the tetralogy's one inverted signature — here the *positive* verdict is the window-witnessable side — and its most complete weld inventory, including the executed negative welds at the conjecture's own boundary. Nothing below asserts the Hodge conjecture, its negation, or its independence; everything below is the proof of what deciding it costs, drawn from the prior work and cited into the tetralogy's common form.

1. Setting

1.1 The statement and the residue lens

Let X be a smooth projective variety over $\mathbb{C}$. The rational Hodge classes in degree $2k$ are $\text{Hdg}^k(X) = H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$; the Hodge conjecture (HC) asserts every such class is a $\mathbb{Q}$ -combination of classes of algebraic subvarieties. For a class α , the Compiled Edition's residue lens records the open predicate: $\mathcal{J}(\alpha) = 0 \iff \alpha$ is algebraic. By standard spreading-out, HC is equivalent to its restriction to varieties over $\mathbb{Q}$ — countable, effectively presentable data (Compiled Edition §25.1).

1.2 The frames

Definition 1.2 (Hodge frames). A *frame* is a finite station of any of the following circumscription continua: enumeration of candidate algebraic cycles over $\mathbb{Q}$ of bounded degree, height, and presentation complexity, with class comparison to bounded precision (the *search continuum*); verification that a target class passes a finite battery of necessary conditions — rationality, (k,k) -type, harmonicity, absolute-Hodgeness, motivated-ness (the *filter continuum*); and closure of HC on specified loci — divisor classes, particular families, particular strata (the *locus continuum*). A *window observable* is any functional of such a station. The current record stations are Tier-I facts of the Compiled Edition: all filters passed by every surviving target (Main Theorem, component B); the fourfold Weil stratum, the generalized-Kummer Kuga–Satake locus, and countably many K3 families closed affirmatively (§22); the open frontier located at the Weil sixfolds of discriminant $\neq -1$, sharpest member the McMullen rigid family (§23).

Definition 1.3 (Frame-internal decidability; weld). As in T1–T3 verbatim.

2. Classification: the inverted asymmetry

Lemma 2.1 (Logical form; Tier I). For a fixed target (X, α) over $\mathbb{Q}$: the positive verdict $\mathcal{J}(\alpha) = 0$ is Σ -shaped over an effectively enumerable object space — witnessed, if true, by a single finite object (a cycle with a verification certificate) arising at some station of the search continuum. The negative verdict $\mathcal{J}(\alpha) \neq 0$ is Π -shaped: non-membership over the entire infinite cycle enumeration, witnessed by no station and certifiable only by a separating invariant. The full conjecture is a Π -statement over all instances of these Σ -statements.

Proof. Compiled Edition, Main Theorem component D (witness asymmetry), restated in the tetralogy's vocabulary; the enumeration and the finite presentability of witnesses are the spreading-out facts of §1.1. ■

Theorem 2.2 (One-sided frame, inverted; Tier I/II). The Hodge search continuum can confirm instances but can never refute one: every frame-internal activity is, logically, witness search. This is the exact mirror of the Riemann case (T2, Theorem 2.2), where the frame could refute and never confirm. The two problems are the two chiralities of the same structure: a one-sided circumscription continuum facing a tail on its blind side. Consequently the Hodge record — every filter passed, loci closing affirmatively decade after decade, no refutation ever produced — is exactly the output the logical form predicts, and carries no verdict information for the open locus beyond the witnesses it has so far failed to find and the refutations it is structurally incapable of producing.

Proof. The first sentence is Lemma 2.1. The incapacity clause: a refutation requires certifying Π -non-membership, which by Lemma 2.1 needs an invariant; the filter continuum's members are all *necessary* conditions of algebraicity, satisfied by the targets (Main Theorem B), hence non-separating by construction — this is Theorem 17.6 of the Compiled Edition, cited below as the certification prong of B''' . ■

Remark 2.3 (The window is treacherous here by executed refutation; Tier I). As in T2 (Littlewood) and T3 (triviality; abelian deconfinement), the domain exhibits its own proof that window extrapolation fails — and here the exhibits are *executed refutations at the conjecture's boundary* (Compiled Edition §36, the nil cases):

1. **Integral coefficients:** the integral Hodge conjecture is false — Atiyah–Hirzebruch (torsion/Steenrod filters) and Kollár (degree obstructions). At the integral boundary the separating invariant *existed*, and fired. Boundary theorem: the obstructions are torsion and divisibility phenomena, vanishing identically over $\mathbb{Q}$ — the filters that killed the neighbor provably cannot cross into the rational conjecture.
2. **The original general formulation:** false; refuted and corrected by Grothendieck.
3. **The Kähler extension:** false; Voisin's compact Kähler counterexamples. Boundary theorem: projectivity is load-bearing.

The surviving conjecture is the theorem-lined residue of three successful attacks — the strongest possible form of the warning that unanimity of window evidence (here: unanimity of filter passage) does not extrapolate, since immediately across each boundary the same style of evidence coexisted with refutation.

3. The obstruction theorems: the identity with the Compiled Edition

The tetralogy's B/C/D theorems for Hodge are not constructed here; they are **identified** — each is a theorem already proven in the prior work, now recognized as the circumscription theorem it is. This section is the paper's central claim made rigorous: same structural obstruction, definitively.

Theorem B''' (Frame-blindness — the instrument theorems)

Theorem B''' (Tier I; = Compiled Edition Theorems 17.4, 17.6, 18.2, 20.1; C1 Theorems 4.3–4.4). Two prongs, as in T3:

1. **(Construction prong.)** No conjugation-invariant reattachment — no operation of the declared instrument class $\mathfrak{Q}$, under any finite or infinite iteration of the reframing monoid — produces a cycle class in the conjugation-anti-invariant target component: the eigenspace obstruction gives the uniform quantitative bound $O(r, \alpha) \geq \|\alpha\|$ for every $r \in \mathfrak{Q}$, whence $I_{\mathfrak{Q}}(\alpha) = \|\alpha\| > 0$ with the apparent obstruction vanishing identically, $A \equiv 0$ — iteration provably semantic. This is the exact Hodge form of T1's Theorem B and T3's kinematic- non-abelianness bar: hypotheses invariant along the orbit cannot decide a verdict the orbit does not decide.
2. **(Certification prong.)** No certificate whose acceptance is implied by Hodge membership separates algebraic from non-algebraic within the Hodge locus: the induced partition is trivial (Theorem 17.6). This is the exact Hodge form of T2's Theorem B': the counterfeit role played there by Davenport–Heilbronn is played here by the Hodge locus itself — every filter in the continuum is passed by all candidates, deciders and non-deciders alike, so filter-frame arguments are barred by construction rather than by exhibited counterexample. Where an exhibited counterexample does exist — the integral boundary — the filter fired and the conjecture *fell*, confirming that the bar is about this frame, not about refutation in general.

Proof. Citations as stated; the identification of the two prongs with the T-series forms is definitional once Definitions 1.2–1.3 are fixed. ■

Theorem C''' (Windows buy windows — searches, filters, loci)

Theorem C''' (Tier I/II). (i) Cycle search to any bound decides nothing negative (Lemma 2.1) and, positively, decides only the instances it closes. (ii) Filter verification to any depth is non-separating (B''' prong 2). (iii) Locus closures relocate the frontier without deciding the general statement — the Compiled Edition's entire node history (Lefschetz (1,1) closing degree two; Markman closing the fourfolds; the Kummer/K3 closures) is the rigorous record of exactly this: each closure is a genuine weld *for its locus* and a coordinate change for the residue (the three-shape frontier of V4 §10, the migration ledger of §22–23). (iv) The analogue of T1's Theorem C and T2's C' (ii): an *effectivity theorem* — a computable bound f with " α algebraic $\implies$ realized by a cycle of degree/height $\leq f(\text{inv}(X, \alpha))$ " — would render every instance decidable by a terminating search and is therefore not a lemma toward the problem but a decision procedure for it; no such bound is known, and the one place a family-specific finite reduction has been achieved — the McMullen rigid family, whose existence question is reduced to a finite computation — is correctly understood as a *delivered partial C''' weld*, the Hodge analogue of Chen–Hou's machine-certifiable compactification, pending execution.

Proof. (i)–(iii) are the cited prior results read through Definition 1.2. (iv): the hypothesis of such a theorem is checkable by bounded enumeration with certified class comparison over $\mathbb{Q}$; the theorem plus the computation outputs the instance's verdict either way. ■

Theorem D''' (The escape routes — the verdict lattice and the forcing wall)

Theorem D''' (Tier I/II; = Compiled Edition §§24–25, Main Theorem C–E). (i) (*Witness asymmetry, inverted.*) The positive verdict admits a finite witness; the negative requires an invariant strictly finer than *motivated* — an object of which no instance exists, whose construction would refute the standing conjectural architecture of the field, and whose executed exemplars (the torsion filters) provably vanish rationally. (ii) (*The forcing wall.*) Under the spreading-out formalization, the Hodge statement sits at Shoenfield-absolute complexity; forcing cannot alter its truth value; independence-via-forcing is barred — the conditional stated and discharged as in the prior work. (iii) (*The four-cell lattice.*) The demand "unprovable $\implies$ provably unprovable" is refuted with the exhibited witness $\text{Con}(\text{ZFC})$; the open cell is ineliminable by theorem; and no consistency-strength attachment for HC exists or is indicated. The metamathematical route is walled on the one lane and priced at the invariant of (i) on the other.

Proof. Citations as stated (Compiled Edition Theorems 24.1–24.3, Corollary 25.2, Main Theorem components C–E). ■

Corollary F''' (The circumscription obstruction, cycle-theoretic form)

Corollary F''' (Tier II). The Hodge verdict is not frame-internally decidable: not witnessed negatively by any station (Lemma 2.1, Theorem 2.2), the record being structurally incapable of refutation; not decided by any instrument operation under any iteration, the apparent obstruction being identically zero (B''' prong 1 — the strongest "iteration is semantic" identity in the tetralogy, computed rather than argued); not decided by any filter at any depth (B''' prong 2); not decided by searches, filters, or locus closures except locus-wise (C'''); and not outflanked metamathematically (D'''). Every decision factors through a weld — and in this domain the claim is not merely proven but *historically instantiated in both directions*: affirmative welds have closed strata (Lefschetz, Markman, Kummer/K3), a negative weld has closed the integral boundary (Atiyah–Hirzebruch), and the surviving open locus is exactly the region no existing weld reaches. The station at which one stops searching, filtering, or closing loci and pronounces on the remainder is, absent a new weld, arbitrary. ■

4. The weld taxonomy, instantiated — the weld-manufacturing domain

Hodge theory holds a distinguished place in the taxonomy: it does not merely *consume* welds; it **manufactures the positivity mechanisms the other domains import**. The Castelnuovo–Severi inequality by which Weil proved RH for function fields (T2 §4) is the Hodge index theorem on a surface; the (W3') component priced there is an arithmetic Hodge index theorem. The taxonomy below is therefore also a genealogy.

Type I — Criteria and reformulation welds

- **Lefschetz (1,1):** the paradigm criterion weld of the entire tetralogy — a window-checkable condition (integral (1,1)-type) coupled by theorem to existence of the witness (a divisor). It decides its locus *completely*: degree-two Hodge classes are closed, everywhere, forever. The exhibit that frame-internal decision is possible exactly where, and only where, a weld exists.
- **The residue lens and the reformulation family:** $\mathcal{I}(\alpha)$, the verdict lattice, the door specifications — theorem-grade relocations of the open predicate (Compiled Edition §2.2, §14, Remark 20.3), the exact analogues of the Lagarias/Nyman–Beurling family in T2.

Type II — Positivity welds

- **The Hodge–Riemann bilinear relations and the Hodge index theorem:** the domain's native positivity structure — the mechanism inside every Lefschetz-type argument, and the export that decided RH over function fields. The domain that poses the tetralogy's hardest surviving cycle question is the same domain that supplies the positivity welds by which its siblings' questions fell: the structural reading's most striking internal cross-link.
- **Deligne's absolute-Hodge theorem and André's motivated cycles:** filter-strengthening welds — each couples the targets to a finer necessary structure, narrowing where a refuting invariant could live and, by Main Theorem B, emptying every existing candidate.

Type III — Carrier welds: the migrations, in both directions

- **Markman (2022–), with Floccari, Fu–Floccari, O'Grady, Voisin, Varesco:** the affirmative carrier weld executed on this problem's own frontier — carrier change to hyper-Kähler moduli, deformation-stable hyperholomorphic and semiregular sheaf classes, transport and specialization — closing the entire Weil fourfold stratum (all fields, all discriminants) and the generalized-Kummer Kuga–Satake locus. The event carries the tetralogy's unique evidentiary weight: the Compiled Edition's instrument had proven the frontier frame-internally undecidable and posted the door's exact specification; the crossing occurred through that door, by that weld type, at those coordinates. The circumscription analysis predicted the shape of its own supersession and was confirmed by publication record.
- **Kuga–Satake:** the structure-level carrier — a genuine weld of Hodge structures whose algebraicity residue relocates rather than decides (V4 §8), the clean exhibit of Type III's anatomy separated into its two halves.
- **Atiyah–Hirzebruch / Kollár:** the *negative* carrier welds executed at the integral boundary — the tetralogy's only domain where the refutation fate has actually been achieved against a Hodge-type statement, with the boundary theorems (torsion vanishing rationally) marking exactly why the achievement terminates at the rational core.
- **The pending instance:** the McMullen rigid family's finite computation — the domain's Chen–Hou-shaped, machine-certifiable compactified tail, delivered as a reduction and awaiting execution.

5. The spine, closed

With T4 the table of T2 §5 and T3 §5 completes over all four domains; the Hodge column reads: two coupled ladders — the algebraic-cycle ladder and the transcendental Hodge-theoretic ladder, coupled by the cycle class map; verdict's logical form — Σ -witnessable instances under a Π aggregate, the inverted chirality of Riemann; frame-internal bar — Theorems B^{'''}–D^{'''}, which are the Compiled Edition's instrument completion, non-separation, quantifier exhaustion, verdict lattice, and forcing wall under their tetralogy names; window record, verdict-silent — every filter passed, loci closing affirmatively for decades, structurally incapable of refutation; weld record — the richest in the tetralogy: affirmative carrier welds executed (Markman era), a negative weld executed at the boundary (integral case), the criterion paradigm (Lefschetz (1,1)), the positivity mechanisms exported to the sibling domains, and one pending machine-certifiable instance.

The identity, stated once (Tier II). The four instantiations are one theorem: *a verdict posed as an exact global property of the seam between two coupled descriptive ladders is decidable by no station of the seam's native circumscription continuum; the continuum's one-sidedness (which side can witness) is fixed by the verdict's quantifier chirality — Π -confirmation-blind for Riemann, Σ -refutation-blind for Hodge, doubly tail-blind for Navier–Stokes and Yang–Mills; the domain's own landmark theorems bar every frame-internal route; and every recorded decision, in every domain, in both directions, is a weld.* The Hodge case contributes the identity's proof-of-concept in full: both fates executed, the prediction confirmed by a dated event, and the calculus's $A \equiv 0$ identity computed. **The issue is not domain-dependent; it is structural** — the origin statement, now closed over the tetralogy, at exactly its earned strength.

The prime-origin closure, final form (Tier II). The home domain's lesson terminates here in its cleanest algebraic form: each candidate cycle comparison is a completed, decidable fact (the search continuum's stations); the negative aggregate over all of them is the tail; and the one region of mathematics where the aggregate closes absolutely — the primes — is precisely where the object space is finite. Hodge, Riemann, Navier–Stokes, and Yang–Mills are the four canonical ways an infinite object space breaks that closure, and the Rope-and-Sand diagrams, traversed far enough, were always going to arrive at all four.

6. Confirmation rather than correction

Unlike T1–T3, this paper's §6 corrects nothing, because the prior Hodge work was executed under the discipline the tetralogy formalizes — indeed it is where that discipline was forged. The ledger is one of identification:

1. The Compiled Edition's **instrument completion theorem** (20.1) is Theorem B^{'''}; its **uniform separation and $A \equiv 0$ identity** (C1, 4.3–4.4) is the tetralogy's strongest "iteration is semantic" result, the only one computed as an identity rather than derived as a bar.
2. Its **quantifier-exhaustion theorem** (Main Theorem E) is the tetralogy's frame-classification lemma in its most general form: the four readings of "exhaust the answer space," each closed or walled.

3. Its **verdict lattice and Con(ZFC) refutation** (§24) is Theorem D'''' 's metamathematical half, and was the tetralogy's first proof that the demanded three-cell stopping condition is a theorem-violating demand.
4. Its **door pricing** (Remark 20.3, §§26–27) is the (W)-specification pattern of all four papers; and the external-review refinement absorbed in the program's later exchanges — the rigid-point specialization gap as the named sub-door for the sixfold locus — is retained as the current sharpest coordinate of $(W1''')$.
5. Its **historical validation** — the Markman crossing at the posted coordinates — is the tetralogy's single strongest piece of evidence that the structural reading is *true*, not merely consistent: the analysis was falsifiable in real time, and was confirmed.

7. The priced specification, restated in tetralogy form

(W1''') **The carrier:** hyper-Kähler / moduli embedding for the sixfold targets that evades the rigidity obstruction — the rigid-point specialization sub-door — or an alternative carrier for the rigid family directly. *Priced by:* Markman (delivered on the fourfold stratum). **(W2''')** **The construction:** deformation-stable algebraic sheaf classes transported and specialized at the rigid points; the McMullen finite computation as the pending machine-certifiable instance. *Priced by:* the Markman-era constructions; Chen–Hou as the cross-domain exhibit of certified finite execution. **(W3''')** **For the negative fate:** an invariant strictly finer than motivated that a surviving target violates. *Priced by:* Atiyah–Hirzebruch and Kollár — executed at the integral boundary — with the boundary theorems standing as the proof of what extension would require. **(W4''')** **The transfer:** the cycle class map and its comparison isomorphisms — proven, classical, complete.

The tetralogy's inversion table thereby closes: Navier–Stokes — transfer cheap, profile missing; Riemann — transfer proven, carrier missing; Yang–Mills — transfer proven, carrier partially built; **Hodge — transfer proven, carrier delivered on the majority stratum, one named family reduced to a finite computation.** Hodge is the furthest along of the four, and the tetralogy's ordering of the problems by distance-to-weld is itself a Tier-II finding of the completed spine.

8. What this paper proves, in one paragraph

That the Hodge verdict is a tail predicate of inverted chirality — positively Σ -witnessable, negatively Π -blind — whose native frames can confirm instances and are structurally incapable of refutation (Lemma 2.1, Theorem 2.2); that the domain exhibits its own window-treachery in executed form, the conjecture's three boundary refutations standing as nil cases with theorem-lined non-extension (Remark 2.3); that the frame-internal bars are the Compiled Edition's instrument theorems under their tetralogy names — the eigenspace obstruction with uniform constant and the $A \equiv 0$ identity as the construction prong, non-separating certification as the filter prong, effectivity as the price of a search criterion, and the verdict lattice plus forcing wall as the metamathematical closure (Theorems B'''' – D'''' , by identification with the prior work); that the weld taxonomy is instantiated here more completely than anywhere in the tetralogy — criterion, positivity, and carrier welds all executed, in both directions, with the domain additionally serving as the manufacturer of the positivity mechanisms its sibling problems import (§4); that the Markman event constitutes the structural reading's real-time falsifiable confirmation

(§6); and that the completed spine identifies one structural obstruction across all four domains, orders them by distance-to-weld with Hodge nearest, and thereby earns, in full and at no more than its strength, the origin statement: the posing generates the outcome, and the issue is structural.

9. Standing and falsification surfaces

Asserted as proven: Lemma 2.1, Theorems 2.2, B''' , C''' , D''' , Corollary F''' — each by identification with, and citation of, theorems proven in the Compiled Edition V4–V8 and C1, at the tiers there declared; the taxonomy classifications and the spine identity (Tier II over the Tier-I rows); the specification's necessity relative to §3–§4.

Not asserted: HC, $\neg$ HC, their independence; the achievability of $(W1''')$ – $(W3''')$ on the open locus (delivered on closed strata; open on the sixfolds); anything beyond the Compiled Edition's own standing, which this paper inherits together with its falsification schema R1–R8.

To defeat this paper, exhibit one of:

- **$F1'''$** — an error in any identified theorem (equivalently, a successful exhibit under the Compiled Edition's R1–R8);
- **$F2'''$** — a frame-internal refutation of an HC instance (a negative verdict from searches or existing filters — refutes Theorem 2.2's incapacity clause);
- **$F3'''$** — an effectivity bound not equivalent to instance-decidability (refutes C''' (iv));
- **$F4'''$** — a rigorous cycle-existence verdict outside the three-type taxonomy;
- **$F5'''$** — a valid resolution of the open locus lacking both $(W1''')$ – $(W2''')$ and $(W3''')$.

Absent such an exhibit, the tetralogy stands complete and the origin statement stands proven at its fourth station and over the whole: *the Hodge verdict is not frame-internal; its continuum can only ever confirm, never refute; every decision it has ever received, affirmative or negative, came through a weld; the analysis that says so was tested against a live frontier and confirmed by the crossing it predicted; and the one structural obstruction, now exhibited in fluid dynamics, arithmetic, quantum field theory, and algebraic geometry under a single falsifiable form, is the thing the traversals were tracking from the first diagram in the sand.*

References and source anchors

Program corpus (primary; this paper is their identification). Semita, L., *The Hodge Traversal, Compiled Edition V4–V8* (instrument theorems 17.4/17.6/18.2/20.1; quantifier exhaustion; verdict lattice §24; forcing wall §25; Main Theorem A–E; nil cases §36; genre sweep §35; falsification schema R1–R8); *The Obstruction Calculus, Consolidated (C1)* (uniform separation, $A \equiv 0$, the Markman monotonicity witness); *The Residue Thesis (R1)*; *Tail Predicates and the Circumscription Obstruction (T1)*; *The Critical Line as a Tail Predicate (T2)*; *The Mass Gap as a Tail Predicate (T3)*; *The Rope-and-Sand Gambit* (PRIMES repository).

Classical and modern Hodge theory. Lefschetz (1,1); Hodge–Riemann bilinear relations; the Hodge index theorem; Deligne, *Hodge cycles on abelian varieties*; André, motivated cycles; Weil, exotic classes on abelian fourfolds; Moonen–Zarhin; Kuga–Satake; Schoen, van Geemen.

The Markman-era closures. Markman (2022–), algebraicity of Hodge–Weil classes on abelian fourfolds, all fields and discriminants; Floccari; Fu–Floccari; O’Grady; Voisin; Varesco; Schnell (survey). The McMullen-curve rigid sixfold family (2026), absolute-Hodge, inaccessible to known methods, reduced to a finite computation.

Boundary refutations. Atiyah–Hirzebruch (1962); Kollár; Grothendieck (1969), the corrected general Hodge conjecture; Voisin (2002), Kähler counterexamples.

Cross-domain exhibits. Weil (1948) and Deligne (1974) — the Hodge index mechanism deciding RH over function fields (the taxonomy’s genealogy); Chen–Hou — certified finite execution as the $(W_2^{(m)})$ pattern; Perelman, Elgindi — the tetralogy’s other carrier exhibits.

Logic. Gödel; Shoenfield; the Con(ZFC) witness of the verdict lattice, as in the Compiled Edition §24.